

Java FSD - Day 9 Summary

Hollow Pattern Printing Using For Loop

- Day 9 focused on creating hollow patterns using nested for loops.
- In hollow patterns, stars are printed only on the border and spaces are printed inside.

Hollow Square Pattern

- A hollow square pattern prints stars on the first row, last row, first column, and last column.
- The inner area is left empty by printing spaces.
- This pattern helps us understand row and column conditions.

Hollow Rectangle Pattern

- A hollow rectangle works similar to a hollow square but the number of rows and columns can be different.
- Stars are printed only on the boundary of the rectangle.
- Spaces are printed inside the rectangle.

Common Logic Used

- Print a star (*) when the row is first or last.
- Print a star (*) when the column is first or last.
- Otherwise print a space.

Additional Hollow Pattern Questions

- Hollow Inverted Right Triangle
- Hollow Pyramid Pattern
- Hollow Inverted Pyramid Pattern
- Hollow Diamond Pattern
- Hollow Number Square Pattern
- Hollow Right-Aligned Triangle Pattern
- Hollow Hourglass Pattern

Important Points

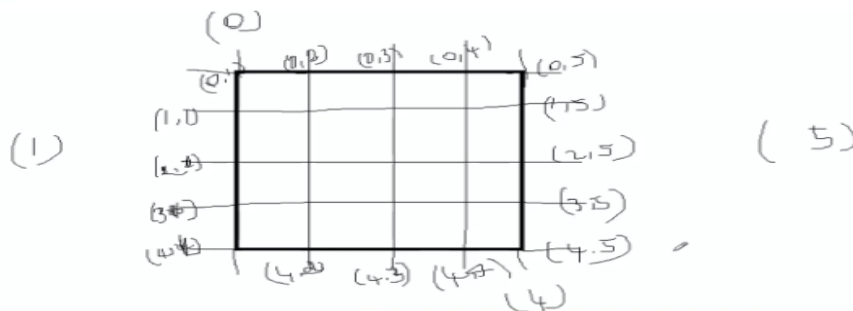
- Nested loops are required for hollow patterns.
- Conditions play an important role in deciding where stars should be printed.
- These pattern questions improve logical thinking and coding skills.
- Pattern-based programs are commonly asked in coding interviews.

Reference Images

Hollow Pattern Printing Questions Using for Loop

1 Write a Java program to print a **hollow square star pattern**.

```
public class HollowPattern1 {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n; j++) {
                if (i == 1 || i == n || j == 1 || j == n)
                    System.out.print("* ");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```



Hollow Rectangle Star Pattern

```
public class HollowRectangle {
    public static void main(String[] args) {
        int rows = 4, cols = 7; // rectangle size
        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= cols; j++) {
                // Print stars on border, space inside
                if (i == 1 || i == rows || j == 1 || j == cols)
                    System.out.print("* ");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```



Hollow Pattern Printing Questions Using for Loop

- 4 Write a Java program to print a **hollow inverted right triangle star pattern**.
- 5 Write a Java program to print a **hollow pyramid star pattern**.
- 6 Write a Java program to print a **hollow inverted pyramid star pattern**.
- 7 Write a Java program to print a **hollow diamond star pattern**.
- 8 Write a Java program to print a **hollow number square pattern**.
- 9 Write a Java program to print a **hollow right-aligned triangle pattern**.
- 10 Write a Java program to print a **hollow hourglass star pattern**.